Subject: ACS 547, Noise Control Applications - Module 3 Assignment

Date: March 31, 2025 (Submitted)

Problem 1 - Washing Machine Mount Design

The Matlab code for this problem is listed in Appendix 1.

Problem 1a

Equation calculates the force produced by the imbalanced 1kg mass at a 0.5 m radius from the center of the washing machine as a function of angular velocity,

$$F = m \cdot \omega^2 \cdot r \tag{1}$$

where m is 0.5 kg, ω is the angular velocity in $\frac{\text{radians}}{\text{s}}$, and r is 0.5 m.

Figure 1 shows the force produced by the 1 kg load as a function of angular velocity in RPM.

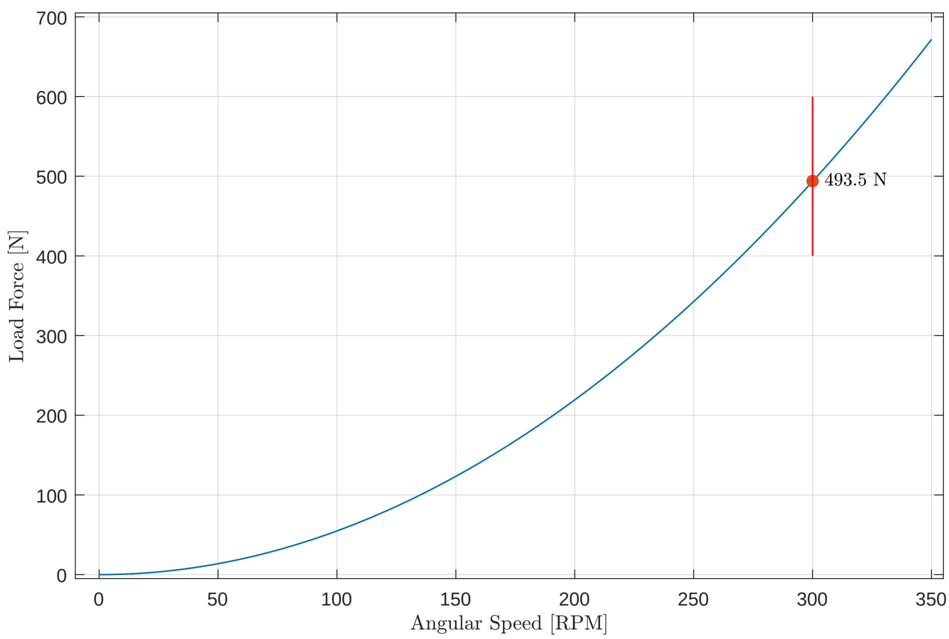

Figure 1: Load force versus washing machine angular speed.

At 300 RPM, or 5 Hz, the imbalanced load force amplitude is 493.5 N.

Problem 1b

The required spring stiffness, k_s, is 9,810 $\frac{N}{m}$. The mount's natural frequency, w_o, is 1.5 Hz.

Problem 1c

Figure 2 shows the transmission loss of the system for a set of damping ratios ranging from 0.001 to 1.

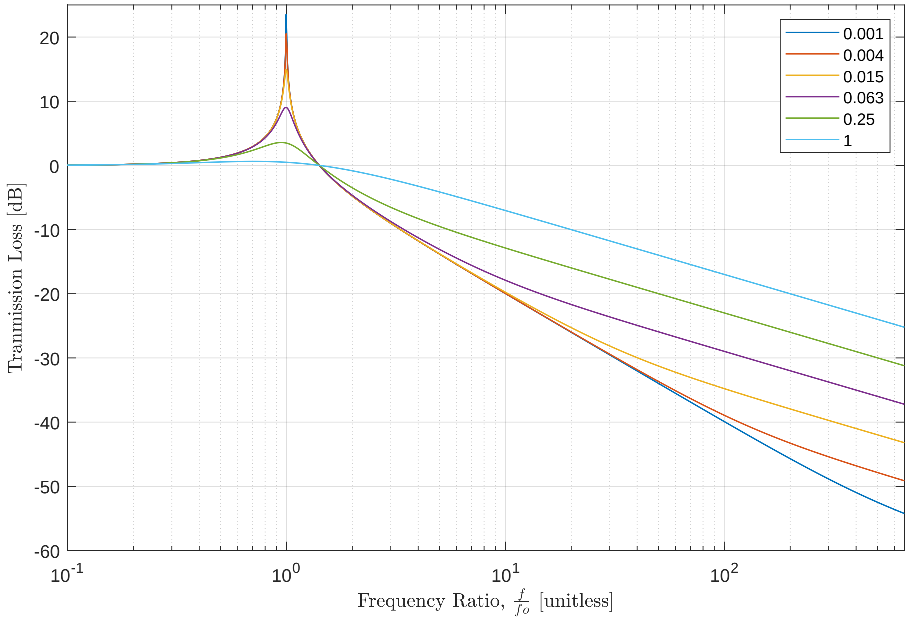

Figure 2: Transmission loss as a function of damping ratio. The natural frequency is 1.5 Hz.

Problem 1d

Figure 3 shows the force applied to the foundation.

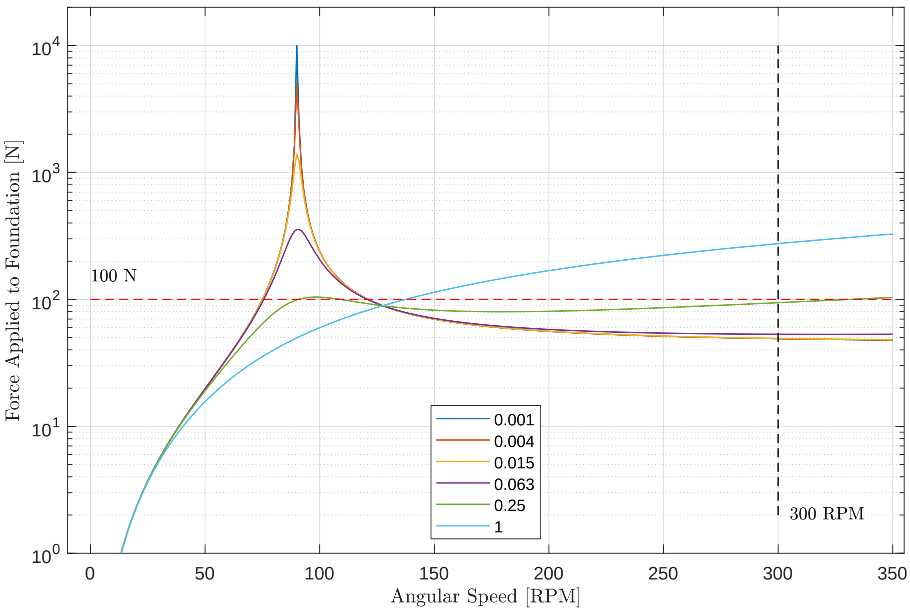

Figure 3: Force applied to the foundation as a function of damping ratio.

No damping ratio makes the applied force less than 100 N across the 0 to 350 RPM operating range.

The best damping ratio is $\zeta = 0.25$. The force applied to the foundation still exceeds the 100 N target in only two small RPM regions: 91 to 109, and 331 to 350 RPM. In both of these regions the force is approximately 104 N.

Problem 1e

The viscous damping coefficient is 519.4 $\frac{\text{kg}}{\text{s}}$ based on a total mass of 110 kg.

Figure 4 shows the admittance.

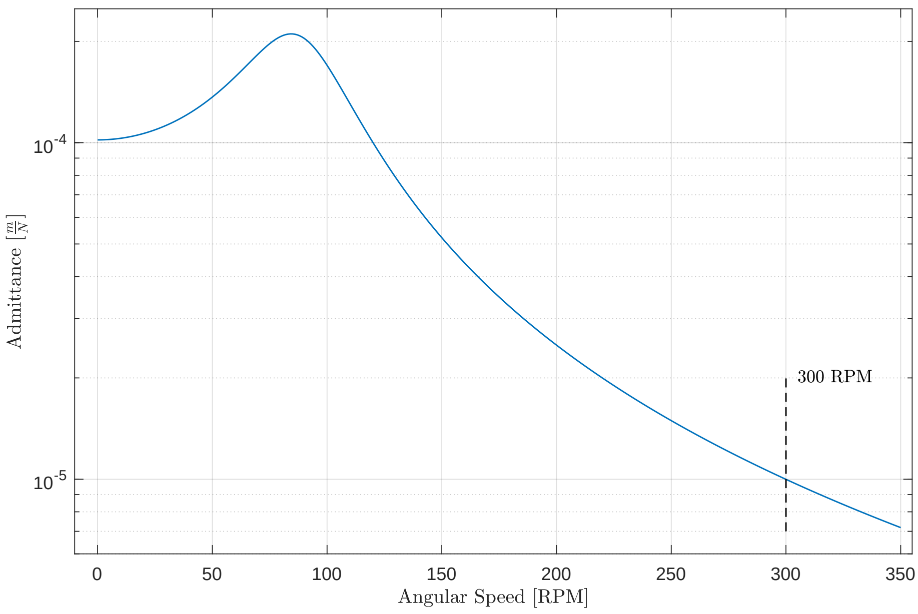

Figure 4: Admittance versus RPM.

The admittance at 300 RPM is 1e-5 $\frac{\text{m}}{\text{N}}$.

Problem 1f

Figure 5 shows the displacement of the washing machine.

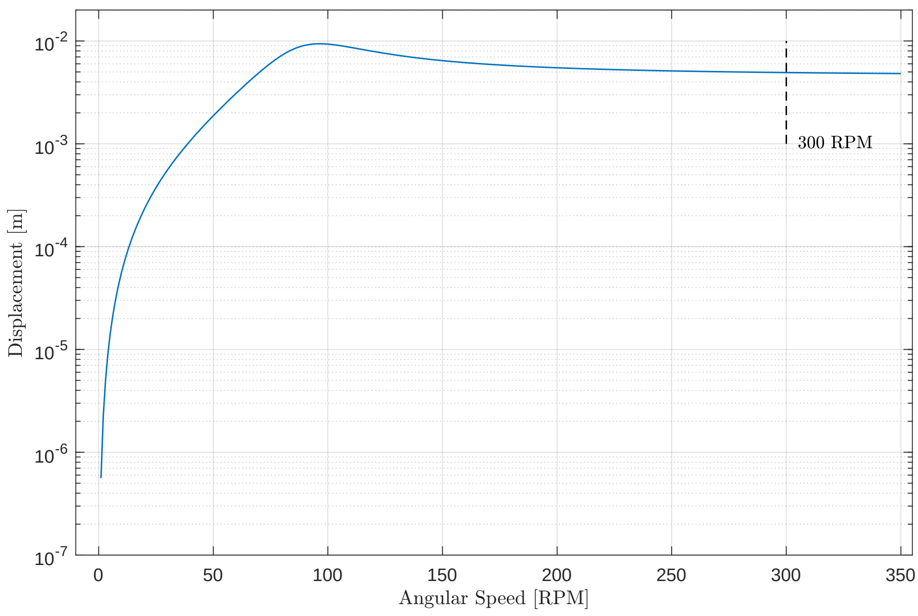

Figure 5: Displacement versus RPM.

The displacement of the washing machine at 300 RPM is approximately 4.9×10^{-3} m. The washing machine undergoes larger momentary displacements as is spins up to 300 RPM, in particular around 100 RPM.

Problem 2 - Washing Machine Mount Design (2 DOF)

The Matlab code for this problem is listed in Appendix 2.

Problem 2a

Figure 1 illustrates the two-stage vibration mount showing the washing machine and the "raft".

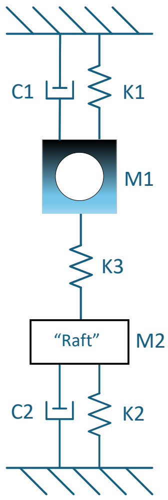

Figure 1: Two-stage vibration mount system for the washing machine.

Problem 2b

Table 1 lists the initial stiffness and damping coefficients for the system.

M1	110	kg
K1	100	$\frac{\mathrm{N}}{\mathrm{m}}$
C1	5	$\frac{\mathrm{kg}}{\mathrm{s}}$[1]

M2	100	kg
K2	100	$\frac{\mathrm{N}}{\mathrm{m}}$
C2	5	$\frac{\mathrm{kg}}{\mathrm{s}}$[2]

M3	0	kg
K3	1,000	$\frac{\mathrm{N}}{\mathrm{m}}$
C3	0	$\frac{\mathrm{kg}}{\mathrm{s}}$[3]

Table 1: Initial values of stiffness and damping coefficients.

Damping was implemented using the damping vector[4].

[1] Element of damping vector.
[2] Element of damping vector.
[3] Element of damping vector.
[4] Not equivalent to using a complex-valued stiffness vector; the imaginary part is the damping for the respective spring

Problem 2c

Figure 2 shows the admittance of the washing machine and the raft.

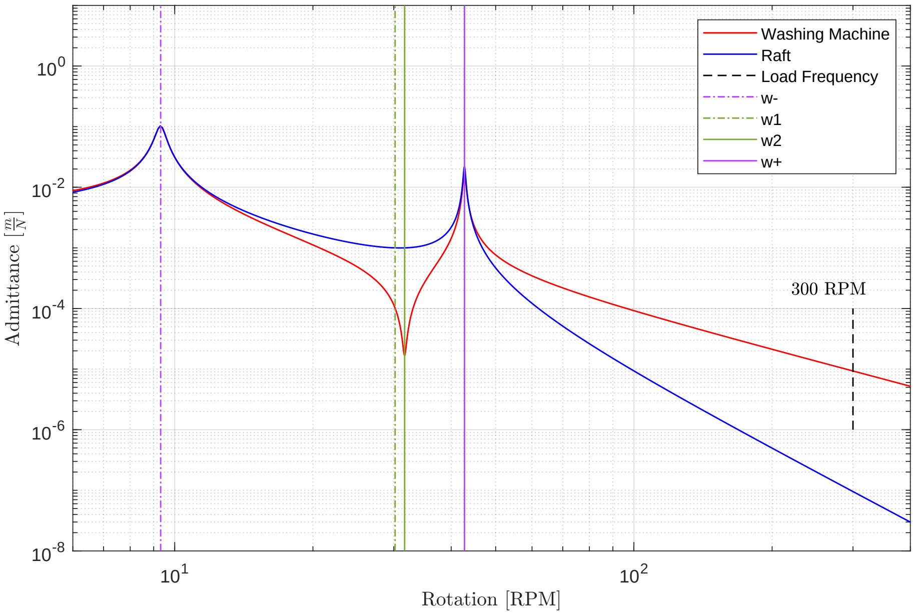

Figure 2: Admittance of the washing machine and the raft with a force on the washing machine.

The admittance of the washing machine is $9.31 \times 10^{-6}\ \frac{\mathrm{m}}{\mathrm{N}}$ at 300 RPM.

Problem 2d

Figure 3 show the displacement of the washing machine and the raft.

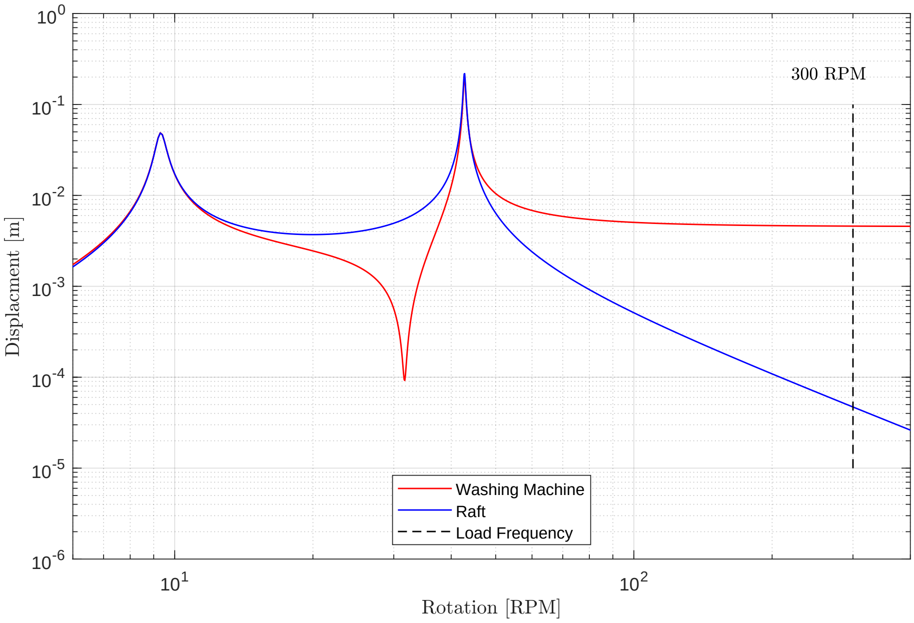

Figure 3: Displacement of the washing machine and the raft with a force on the washing machine.

At 300 RPM, the washing machine displacement is 4.6×10^{-3} m, less than 5.0×10^{-3} m for the first-order system. The displacement of the raft is 4.7×10^{-5} m.

Problem 2e

Figure 4 shows the force applied to the ground by the raft.

The force is is calculated by multiplying the displacement of the raft by the K2 spring constant.

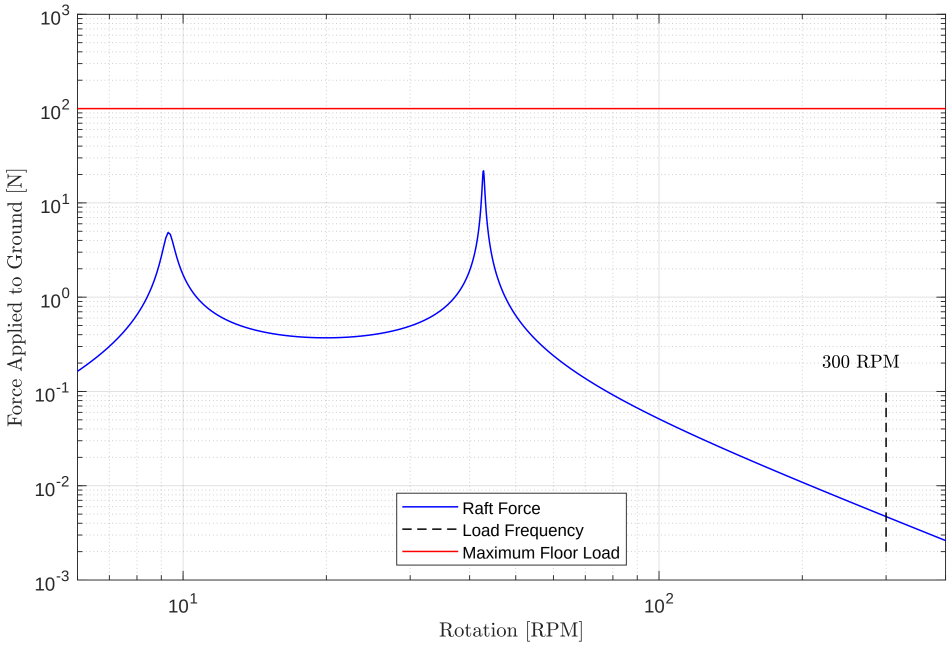

Figure 4: Force applied to the ground by the raft.

The ground force at 300 RPM is 4.7×10^{-3} N. The force on the ground is always less than 100 N.

Problem 2f

The performance of this system was improved by *lowering the stiffness of the coupling spring,* k_3. Lowering k_3 decreases the separation of the two resonance peaks and shifts them down in frequency. This results in lower admittances for the washing machine and the raft at 300 RPM. However, due to the quadratic characteristic of the load force with RPM and the second-order roll-off of the admittance curve for the washing machine, optimization is restricted.

In Figure 3, displacement of the washing machine after the second resonance frequency "plateaus" with much less roll-off relative to the raft displacement. This restricts the degree of optimization that can be done in this case.

Problem 2g

Table 2 compares the admittance, displacement, and ground force for the initial and optimized systems.

	Admittance [$\frac{m}{N}$]	**Displacement [m]**	**Ground Force [N]**
Initial Design	9.31×10^{-3}	4.59×10^{-3}	4.7×10^{-3}
Optimized Design	9.22×10^{-6}	4.55×10^{-3}	4.62×10^{-5}

Table 2: Performance comparison of the initial and optimized designs.

It is apparent that the optimized design greatly reduces the admittance and the ground force. However, the change in displacement is small in comparison.

Problem 3 - Washing Machine Dynamic Vibration Absorber Design

The Matlab code for this problem is listed in Appendix 3.

Part 3a

Figure 1 illustrates the dynamic vibration absorber (DVA) system.

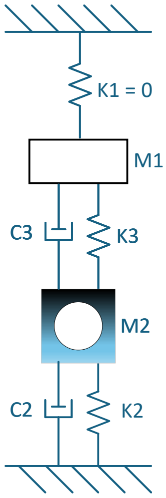

Figure 1: Dynamic vibration absorber system for the washing machine.

Table 3 lists the stiffness and damping coefficients for the system.

M1	15	kg
K1	0	$\frac{\text{N}}{\text{m}}$
C1	0	$\frac{\text{kg}}{\text{s}}$

M2	110	kg
K2	9,810	$\frac{\text{N}}{\text{m}}$
C2	1,962	$\frac{\text{kg}}{\text{s}}$ [5]

M3	0	kg
K3	1e5	$\frac{\text{N}}{\text{m}}$
C3	1e4	$\frac{\text{kg}}{\text{s}}$ [6]

Table 3: Stiffness and damping coefficients for the DVA system.

Damping was implemented by adding with an imaginary component to the respective spring stiffness[7].

[5] 0.2 fraction of respective spring stiffness.
[6] 0.1 fraction of respective spring stiffness.
[7] Not equivalent to using a damping vector in the argument to the given Matlab nDOF function script-file.

Part 3b

Figure 2 shows the admittance response of the original system and DVA system based on the equations presented in the tutorial lecture on DVA system analysis.

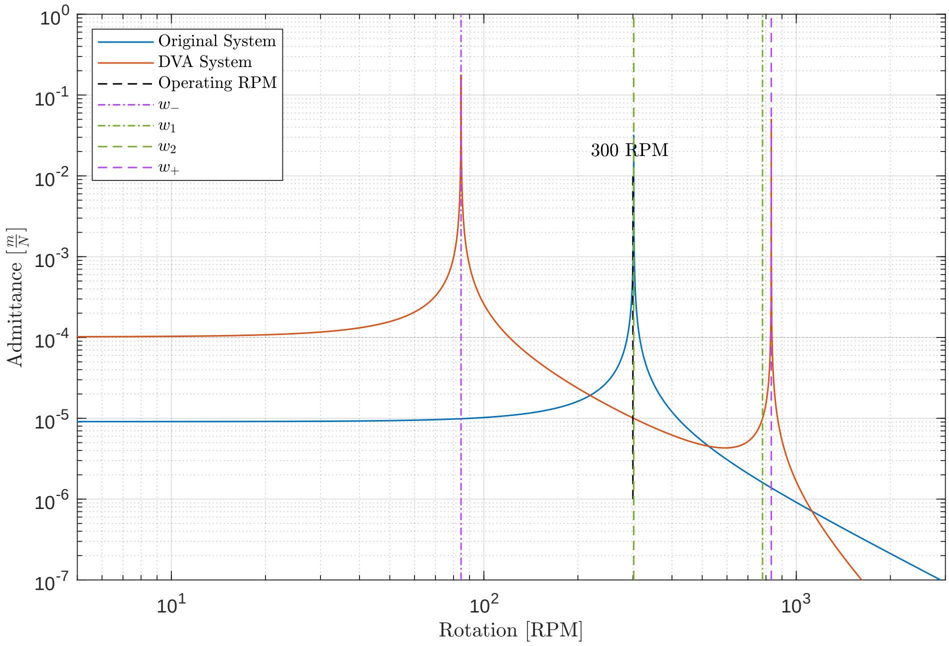

Figure 2: Admittance responses based on the equations from the DVA tutorial lecture.

The washing machine admittance at 300 RPM is reduced from the blue curve resonance peak to a point between the two resonances on the red curve).

Part 3c - Admittance

Figure 3 shows the admittance response of the DVA system using the nDOF function script-file. The stiffness of the coupling spring, k_3, is set to place the null of the admittance response at the 300 RPM operating state of the washing machine. *The admittance at 300 RPM is approximately* 1.2×10^{-6} $\frac{\mathrm{m}}{\mathrm{N}}$.

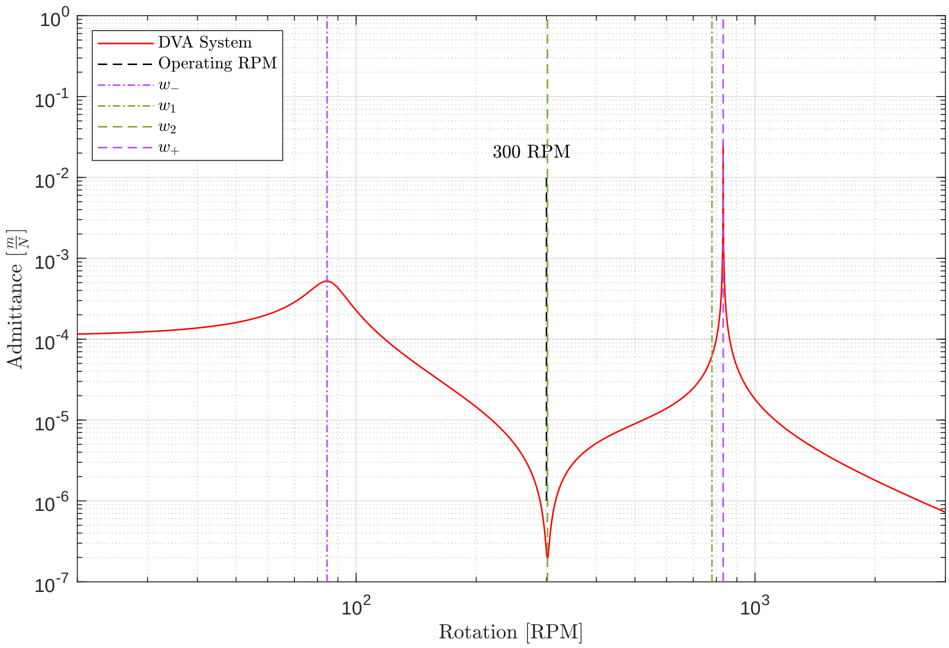

Figure 3: Solution using nDOF script-file function.

Part 3d - Displacement

Figure 4 shows the displacement of the washing machine as a function of RPM. The ramp up to 300 RPM occurs quickly so that the large displacements are transient and the maximum speed is 350 RPM (the large displacement associated with the resonance peak at higher frequencies is not reached). *The displacement at 300 RPM is approximately* 38.2×10^{-6} m.

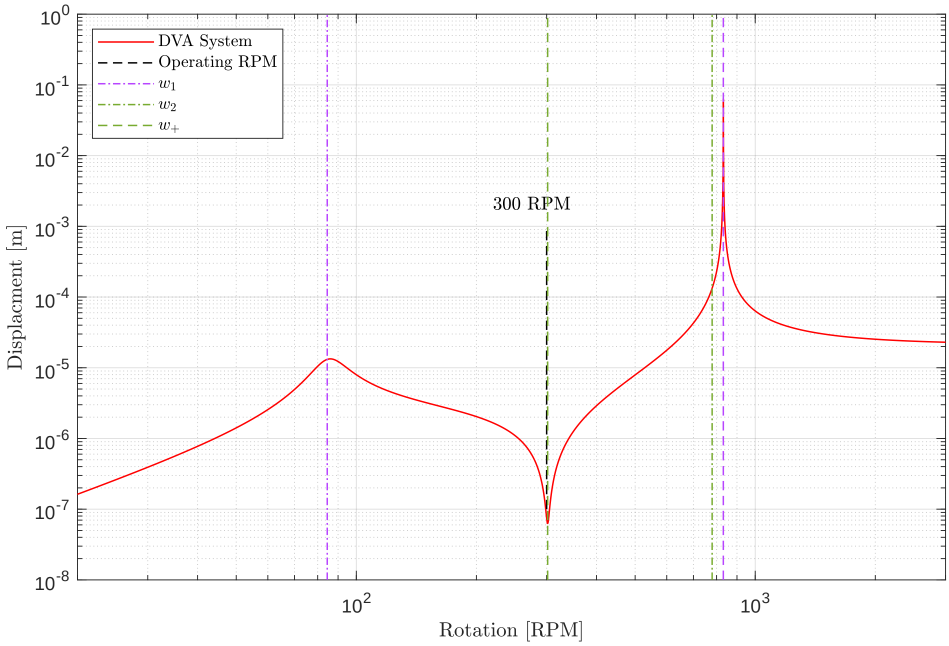

Figure 4: Displacement of the washing machine.

Part 3e

Figure 5 shows the force applied to the floor by the washing machine. *The force applied to the floor at 300 RPM is approximately* 3.75×10^{-3} N.

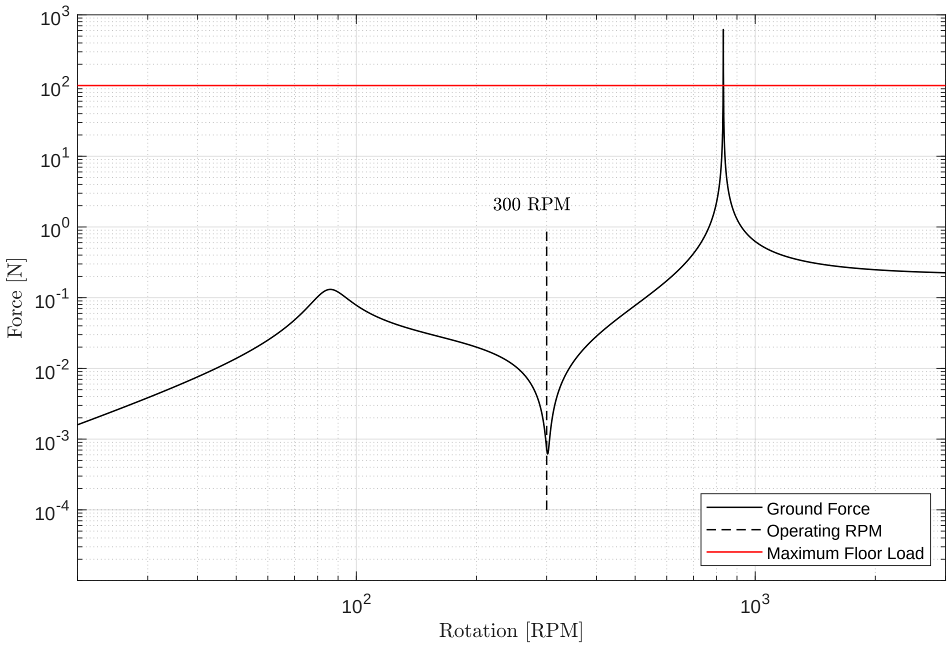

Figure 5: Force applied to the floor.

Problem 4 - Washing Machine Abuse Testing

The Matlab code for this problem is listed in Appendix 4.

Part 4a

Equation 2 is the impulse response for a 1 DOF system.

$$x(t) = \frac{1}{m \cdot w_d} \cdot e^{(-w_o \cdot \zeta \cdot t)} \cdot \sin(w_d \cdot t) \tag{2}$$

where m is the mass of the washing machine, w_d is the damped natural frequency, w_o is the natural frequency, and ζ is the damping ratio.

Figure 1 shows plots of a 1 kg impulse force and the corresponding response of the washing machine.

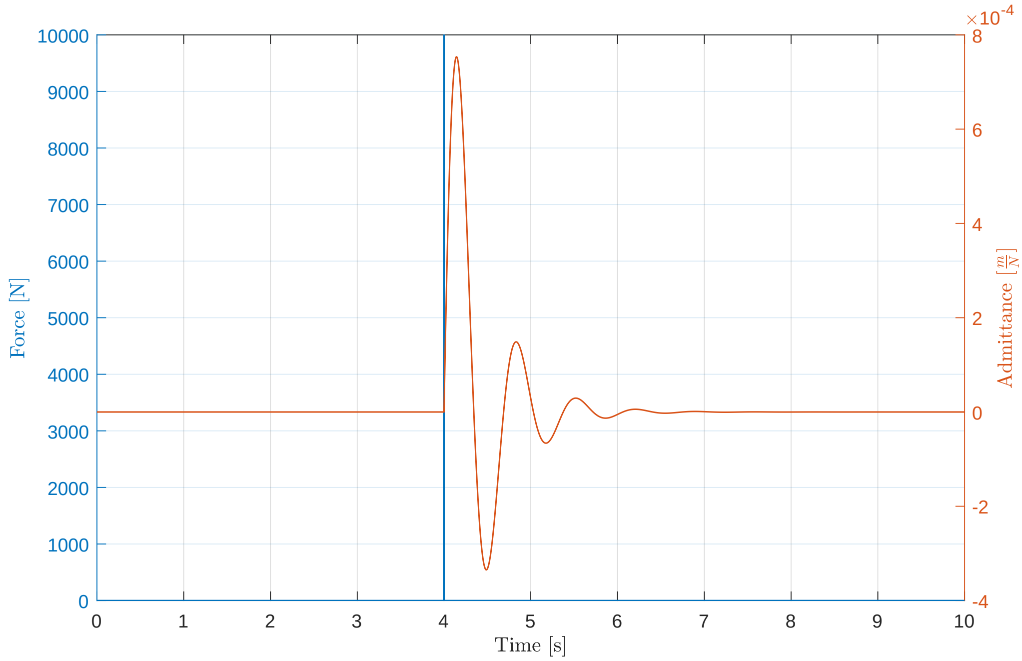

Figure 1: Washing machine impulse response. Here w_o is 9.4 $\frac{\text{radians}}{\text{s}}$, w_d is 9.1 $\frac{\text{radians}}{\text{s}}$, and ζ is 0.25.

Part 4b

A operational sequence for the washing machine created using a sinusoidal forcing function and the impulsive loading with the 5 kg brick. The sequence is as follows,

- Phone body shape will influence the microphone, particularly at high frequencies (?).
- The directionality of internal microphones is not known.
- Calibration of external microphones requires a calibrator and training.
- Cost of external calibrator might be prohibitive (?).
- Potential microphone coupling problems with the external calibrator (?).
- External microphone performance might change with time, handling, and measurement conditions.
- External microphones may not be factually comply with standards (?).
- As of 2016, no smartphone-based sound level measurement solution has met all the electrical and acoustical requirements of the American National Standards Institute (ANSI; 1983) and the International Electrotechnical Commission (IEC; 2013).

Figure 2 shows the response sequence of the washing machine.

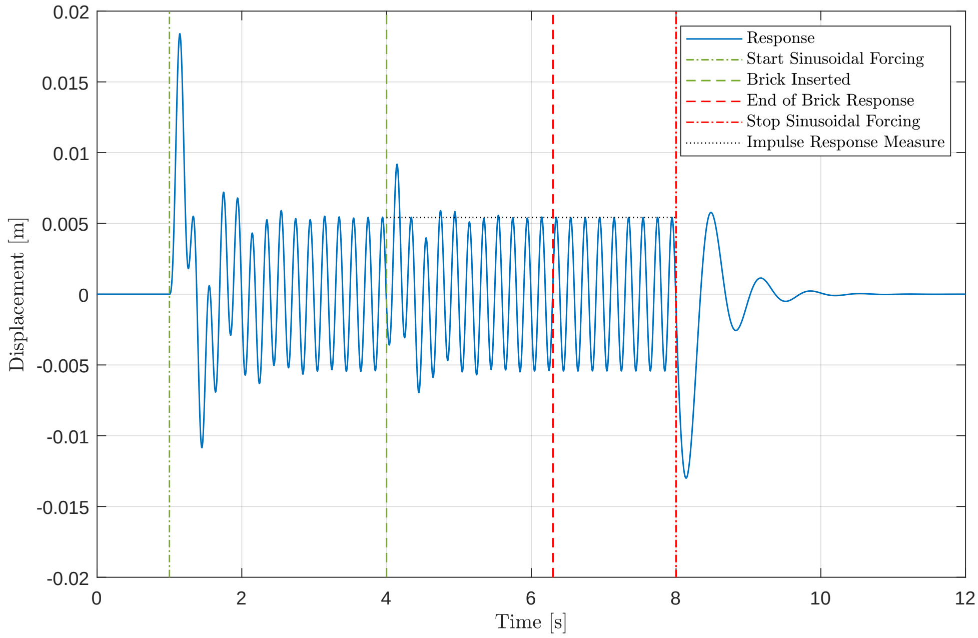

Figure 2: Washing machine response sequence.

Summary

ph

1 Appendix - Matlab Code for Problem 1

```matlab



%% Synopsis

% Problem 1 - Washing Machine Mount Design - 1 Degree of Freedom (DOF)



%% Environment

close all; clear; clc;
% restoredefaultpath;

addpath( genpath( '../00 Support' ), '-begin' );

set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
set( 0, 'DefaultFigureWindowStyle', 'normal' );
set( 0, 'DefaultLineLineWidth', 0.8 );
set( 0, 'DefaultTextInterpreter', 'Latex' );

format LongG;

pause( 1 );



%% Anonymous Function Definitions

h_force = @( force_mass, rotation_speed, load_distance ) force_mass .* rotation_speed.^2 .*
    load_distance;

h_transmissibility = @( epsilon, r ) sqrt( ( 1 + (2.*epsilon.*r).^2 ) ./ ( ( 1 - r.^2 ).^2 + (
    2.*epsilon.*r ).^2 ) );

rpm_conversion_to_radians_per_second = 0.10472;



%% Washing Machine Information

machine.mass = 100; % kg
machine.rotation_speed = 31.4; % radians\s or 300 rotations per minute (RPM)
machine.load_mass = 10; % kg
machine.static_displacement = 1e-2; % m



%% Load Force Information

dynamic_force.mass = 1; % kg
dynamic_force.radius = 0.5; % m

force_frequency = 300 * rpm_conversion_to_radians_per_second / ( 2 * pi ); % 5 Hz

% Maximum force applied to foundation is 100 N.



%% Part 1a

rpm = 0:1:350;
    angular_velocity = rpm .* rpm_conversion_to_radians_per_second; % radians\s


figure( 'Name', 'Problem 1a - Load Force Versus Angular Velocity' ); ...
    plot( rpm, h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ) ); hold on;
    plot( 300, 494, 'Marker', '.', 'MarkerSize', 20 );
    line( [ 300, 300 ], [ 400 600 ], 'Color', 'r' );
    text( 305, 494, '493.5 N' ); grid on;
    xlabel( 'Angular Speed [RPM]' ); ylabel( 'Load Force [N]' );
    axis( [ -10 355 -5 705 ] );



```

```
%% Part 1b

maximum_allowable_displacement = 1e-2;  % m
    ks = ( machine.load_mass * 9.81) / maximum_allowable_displacement;  % 9,810 N\m


total_mass = machine.mass + machine.load_mass;  % 110 kg
    wo = sqrt( ks / total_mass );  % 9.4 radians\s
        fo = wo / (2*pi);  % 1.5 Hz - Natural frequency of the mount.



%% Part 1c

frequency_set = 0.1:0.01:1e3;
    r = frequency_set ./ fo;

damping_ratio = logspace( log10(0.001), log10(1), 6 );


figure( 'Name', 'Problem 1c - Transmission Loss' ); ...
    hold on;
    %
    for index = 1:1:numel( damping_ratio )
        plot( r, 10*log10( h_transmissibility( damping_ratio( index ), r ) ) );
    end
    %
    xlabel( 'Frequency Ratio, $\frac{f}{fo}$ [unitless]' );  ylabel( 'Tranmission Loss [dB]' );
        legend( '0.001', '0.004', '0.015', '0.063', '0.25', '1', 'Location', 'NorthEast' );
    axis( [ 0.1 665  -60 25 ] );
    grid on;  box on;
    set( gca, 'XScale', 'log' );



%% Part 1d

rpm = 0:1:350;  % rotations per minute
    rpm_conversion_to_radians_per_second = 0.10472;  % radians\s
        angular_velocity = rpm .* rpm_conversion_to_radians_per_second;  % radians\s

frequency_set = angular_velocity / (2*pi);
    r = frequency_set ./ fo;


figure( 'Name', 'Problem 1d - Applied Force to Foundation' ); ...
    hold on;
    %
    for index = 1:1:numel( damping_ratio )
        plot( rpm, h_transmissibility( damping_ratio( index ), r ) .* h_force( dynamic_force.mass
            , angular_velocity, dynamic_force.radius ) );
    end
    %
    line( [ 300, 300 ], [ 2 1e4 ], 'Color', 'k', 'LineStyle', '--' );
        text( 305, 2, '300 RPM' );  grid on;
    line( [ 0 350 ], [ 100 100 ], 'Color', 'r', 'LineStyle', '--' );
        text( 0, 150, '100 N' );  grid on;
    xlabel( 'Angular Speed [RPM]' );  ylabel( 'Force Applied to Foundation [N]' );
        legend( '0.001', '0.004', '0.015', '0.063', '0.25', '1', 'Location', 'South' );
    axis( [ -10 355  1 2e4 ] );
    grid on;  box on;
    set( gca, 'YScale', 'log' );



%% Part 1e

% The best damping ratio is 0.25.

C = 0.25 * 2*sqrt( ks * total_mass );  % 519.4 kg\s

% The units of a viscous damping coefficient are Newton-seconds per meter (Ns/m) or kilograms per
    second (kg/s).


m = total_mass;
k = [ ks  0 ];
dampings = [ C  0 ];
```

```matlab
    admittance = nDOF_direct_solution( m, k, dampings, frequency_set, 'admittance' );


figure( 'Name', 'Problem 1e - Admittance' ); ...
    semilogy( rpm, abs( admittance ) );  grid on;
    xlabel( 'Angular Speed [RPM]' );  ylabel( 'Admittance [$\frac{m}{N}$]' );
    set( gca, 'YScale', 'log' );
    %
    line( [ 300, 300 ], [ 7e-6 2e-5 ], 'Color', 'k', 'LineStyle', '--' );
        text( 305, 2e-5, '300 RPM' );  grid on;
    axis( [ -10 355  6e-6 2.5e-4 ] );


% temp2 = abs( admittance );
%     round( temp2( 301 ), 3, 'significant' )



%% Part 1f

displacement = abs( admittance ) .* h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).';


figure( 'Name', 'Displacement' ); ...
    plot( rpm, displacement );  grid on;
    xlabel( 'Angular Speed [RPM]' );  ylabel( 'Displacement [m]' );
    line( [ 300, 300 ], [ 1e-3 1e-2 ], 'Color', 'k', 'LineStyle', '--' );
        text( 305, 1e-3, '300 RPM' );  grid on;
    set( gca, 'YScale', 'log' );
    axis( [ -10 355  1e-7 2e-2 ] );


temp2 = abs( displacement );
    round( temp2( 301 ), 3, 'significant' )



%% Clean-up

if ( ~isempty( findobj( 'Type', 'figure' ) ) )
    monitors = get( 0, 'MonitorPositions' );
        if ( size( monitors, 1 ) == 1 )
            autoArrangeFigures( 3, 4, 1 );
        elseif ( 1 < size( monitors, 1 ) )
            autoArrangeFigures( 2, 2, 2 );
        end
end

fprintf( 1, '\n\n\n*** Processing Complete ***\n\n\n' );



%% Reference(s)
```

2 Appendix - Matlab Code for Problem 2

```matlab



%% Synopsis

% Problem 2 - Washing Machine Two-stage Mount Design



%% Note(s)

% A 2 degree-of-freedom (DOF) system will provide isolation performance 1 DOF system.

% Allowing the top and bottom masses, the washing machine and the raft,
% respectively, to each be attached to ground independently.

% With the force applied to the washing machine and the displacement of the
% washing machine, use FSF( :, 1, 1 ).  However, if the force is applied elsewhere,
% than FSF( :, 2, 2, ) must be used.



%% Environment

close all; clear; clc;
% restoredefaultpath;

addpath( genpath( './00 Support' ), '-begin' );

set( groot, 'DefaultFigurePosition', [ 100  750   750  500 ] ); % x, y, width, height

set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
set( 0, 'DefaultFigureWindowStyle', 'normal' );
set( 0, 'DefaultLineLineWidth', 0.8 );
set( 0, 'DefaultTextInterpreter', 'Latex' );

format ShortG;

pause( 1 );



%% Define Anonymous Functions

h_f_to_rpm = @( f )  ( f*2*pi ) / 0.10471;

h_w_to_rpm = @( w )  h_f_to_rpm( w*2*pi );
h_w_to_f = @( w )  w/(2*pi);

h_rpm_to_f = @( rpm )  ( rpm * 0.10471 ) / ( 2 * pi );


h_force = @( force_mass, rotation_speed, load_distance  )  force_mass .* rotation_speed.^2 .*
    load_distance;



%% Problem 2a

% See report.



%% Problem 2b - Blocked Frequencies of Resonance

% The frequency of the forcing function is 5 Hz and the amplitude is 493.5 N.
fo = 5;  % Hz
    wo = 2*pi*fo;  % 31.4 radians\s


switch ( 1 )

    case 1  % Initial design parameters.

        m1 = 100 + 10;  % kg - Total mass of the washing machine and the load.
```

```matlab
            k1 = 1e2;  % N\m
            c1 = 5;

        m2 = 100;  % kg - UPPER LIMIT OF 100 kg
            k2 = 1e2;  % N\m
            c2 = 5;

        k3 = 1e3;  % N\m -> Larger values produce stronger coupling (higher w+).


    case 2  % Optimized design parameters.

        m1 = 100 + 10;  % kg - Total mass of the washing machine and the load.
            k1 = 1e2;  % N\m
            c1 = 5;

        m2 = 100;  % kg - UPPER LIMIT OF 100 kg
            k2 = 1e2;  % N\m
            c2 = 5;

        k3 = 1e1;  % N\m -> Larger values produce stronger coupling (higher w+).

end


w1 = sqrt( ( k1 + k3 ) / m1 );  % 3.2 radians\s;  washing machine
    f1 = w1 / (2*pi);  % 0.5 Hz
        rpm1 = f1*(2*pi)/0.10471;  % 30.1 RPM

w2 = sqrt( ( k2 + k3 ) / m2 );  % 3.3 radians\s;  raft
    f2 = w2 / (2*pi);  % 0.53 Hz
        rpm2 = f2*(2*pi)/0.10471;  % 31.7 RPM

% Note:  If k3 is set to zero, then these frequencies represent the
% resonance frequencies for each mass on their own.



%% Problem 2b - Coupled Frequencies

mu_4 = ( k3^2 / (m1*m2) );  % 90.9 (radians\s)^4
    mu = ( mu_4 )^0.25;  % 3.1 radians\s

w(1) = 0.5*( ( w1^2 + w2^2 )  +  sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
w(2) = 0.5*( ( w1^2 + w2^2 )  -  sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
    w = sqrt( w );


w_minus = w(2);  %  0.97 radians\s  (lower than w1, 1, and w2, 1.05 radians\s)
    f_minus = w(2)/(2*pi);  %  0.1545 Hz
        rpm_minus = h_f_to_rpm( f_minus );  % 9.3 RPM
%
w_plus = w(1);  %  4.5 radians\s  (higher than w1, 1, and w2, 1.05 radians\s)
    f_plus = w(1)/(2*pi);  %  0.71 Hz
        rpm_plus = h_f_to_rpm( f_plus );  % 42.8 RPM
%
% Note(s):
%
%   The blocked frequencies, w1 and w2, are always between these two frequencies.

clear w mu_4;



%% Problem 2b - Mode Shapes

phi_plus = [ ...
    1; ...
    -(1/mu^2)*sqrt(m1/m2)*(w_plus^2 - w1^2), ...
    ];
%
%   1
%   -1.10

phi_minus = [ ...
    1; ...
    (1/mu^2)*sqrt(m1/m2)*(w1^2 - w_minus^2), ...
    ];
```

```
%
%    1
%    0.99


% Note(s):
%
% When m1 = m2 and k1 = k3,
%
%   a.)  At w_minus, m1 and m2 move the same amount and are in-phase.
%   b.)  At w_plus, m1 and m2 move the same amount, but are out-of-phase.

% In general, for a 2 DOF system there will be 2 modes (in-phase and out-of-phase).

% With different initial conditions (no forcing), the behaviour is a weighted sum of the two modes.



%% Problem 2c

masses = [ m1 m2 ];
stiffnesses = [ k1  k3  k2 ];
dampings = [ c1  0  c2 ];


rpm = 0:0.1:500;  % rotations per minute
    rpm_conversion_to_radians_per_second = 0.10472;
        angular_velocity = rpm .* rpm_conversion_to_radians_per_second;  % radians\s
            frequencies = angular_velocity / (2*pi);  % Hz

FRF = nDOF_direct_solution( masses, stiffnesses, dampings, frequencies, 'admittance' );


admittance = zeros( numel( frequencies ), 1 );

for index = 1:1:numel( frequencies )
    temp = diag( squeeze( FRF( index, 1, 1 ) ) );  % Check indexing to ensure force on washing machine.
        admittance( index ) = abs( temp );
end
%
clear temp temp2;

washing_machine.admittance = admittance;


admittance = zeros( numel( frequencies ), 1 );

for index = 1:1:numel( frequencies )
    temp = diag( squeeze( FRF( index, 1, 2 ) ) );  % Check indexing to ensure force on washing machine.
        admittance( index ) = abs( temp );
end
%
clear temp temp2;

raft.admittance = admittance;


figure( 'Name', 'Admittance' ); ...
    h1 = loglog( rpm, washing_machine.admittance, 'Color', 'r', 'LineWidth', 0.8 );  hold on;
    h2 = loglog( rpm, raft.admittance, 'Color', 'b', 'LineWidth', 0.8 );
    h3 = line( [ 300 300 ], [ 1e-6 1e-4 ], 'Color', 'k', 'LineStyle', '--' );  grid on;
        text( 220, 2e-4, '300 RPM' );
    %
    h4 = line( [ rpm_minus rpm_minus], [ 1e-8 1e1 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '-.' );
    %
    h5 = line( [ rpm1 rpm1 ], [ 1e-8 1e1 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );
    h6 = line( [ rpm2 rpm2 ], [ 1e-8 1e1 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-' );
    %
        h7 = line( [ rpm_plus rpm_plus ], [ 1e-8 1e1 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '-' );
    %
    legend( [ h1 h2 h3 h4 h5 h6 h7 ], 'Washing Machine', 'Raft', 'Load Frequency', 'w-', 'w1', 'w2', 'w+', 'Location', 'NorthEast' );
    %
```

```matlab
    xlabel( 'Rotation [RPM]' );  ylabel( 'Admittance [$\frac{m}{N}$]' );
    axis( [ 6 400  1e-8 1e1 ] );


round( washing_machine.admittance( 3001 ), 3, 'significant' )



%% Problem 2d

dynamic_force.mass = 1;  % kg
dynamic_force.radius = 0.5;  % m


figure( 'Name', 'Displacement' ); ...
    h1 = loglog( rpm, washing_machine.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).', 'Color', 'r' );  hold on;
    h2 = loglog( rpm, raft.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).', 'Color', 'b' );
    h3 = line( [ 300 300 ], [ 1e-5 1e-1 ], 'Color', 'k', 'LineStyle', '--' );  grid on;
        text( 220, 0.21, '300 RPM' );
        legend( [ h1 h2 h3 ], 'Washing Machine', 'Raft', 'Load Frequency', 'Location', 'South' );
    xlabel( 'Rotation [RPM]' );  ylabel( 'Displacment [m]' );
    axis( [ 6 400  1e-6 1 ] );


temp = washing_machine.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).';
    round( temp( 3001 ), 3, 'significant' )

% temp = raft.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).';
%     round( temp( 3001 ), 3, 'significant' )



%% Problem 2e

% Displacement of the raft multiplied by the K2 spring constant.

dynamic_force.mass = 1;  % kg
dynamic_force.radius = 0.5;  % m

h_force = @( force_mass, rotation_speed, load_distance  )  force_mass .* rotation_speed.^2 .* load_distance;

temp = h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).';


figure( 'Name', 'Force Applied to Ground by Raft' ); ...
    h1 = loglog( rpm, raft.admittance.*temp*k2, 'Color', 'b' );  hold on;
    h2 = line( [ 300 300 ], [ 2e-3 1e-1 ], 'Color', 'k', 'LineStyle', '--' );  grid on;
    h3 = line( [ 6 400 ], [ 100 100 ], 'Color', 'r' );
        text( 220, 2e-1, '300 RPM' );
        legend( [ h1 h2 h3 ], 'Raft Force', 'Load Frequency', 'Maximum Floor Load', 'Location', 'South' );
    xlabel( 'Rotation [RPM]' );  ylabel( 'Force Applied to Ground [N]' );
    axis( [ 6 400  1e-3 1e3 ] );


temp = raft.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ).'*k2;
    round( temp( 3001 ), 3, 'significant' )



%% Problem 2f

% return

%% Problem 2g

% return

%% Clean-up

if ( ~isempty( findobj( 'Type', 'figure' ) ) )
    monitors = get( 0, 'MonitorPositions' );
```

```
        if ( size( monitors, 1 ) == 1 )
            autoArrangeFigures( 3, 4, 1 );
        elseif ( 1 < size( monitors, 1 ) )
            autoArrangeFigures( 2, 2, 2 );
        end
end

fprintf( 1, '\n\n\n*** Processing Complete ***\n\n\n' );



%% Reference(s)



%% Comment(s)



%% Regions of Interest

% 5 regions of interest:

% 1.)  Low frequency:  w < w_minus:
%
%   Stiffness controlled;  two masses moving together;  strong coupling (k3 high).


% 2.)  Force at a resonance frequency:  w = w_minus (lowest resonance frequency):
%
%   At lowest resonance frequency;  at w_minus mode, masses moving in same direction (in-phase) with same high amplitude


% 3.) Forcing frequency between w_minus and w2


% 4.)  Forcing frequency w = w2


% 5.)  Forcing frequency w < w < w+
%
%   No a special frequency.


% 6.)  Forcing frequency w = w+
%
% At highest resonance frequency;  at w_plus mode, masses moving in opposite directions (out-of-phase) with same high amplitude


% 7.)  Forcing frequency w+ < w
%
% Mass controlled region;  mass 1 (washing machine) moves more than mass 2 (raft);
```

3 Appendix - Matlab Code for Problem 3

```matlab


% To do:

% remove comment statements

% verify indices for required values



%% Synopsis

% Problem 3 - Washing Machine Dynamic Vibration Absorber (DVA) Design



%% Note(s)

% At 300 RPM, the load frequency, this system will isolated vibration more
% than the 2 DOF and 1 DOF systems.

% The DVA behaves like a Helmholtz resonator, working as a mechnical notch
% filter at a particular frequency.

% DVAs work well in systems that have one dominate mode.  A good
% example of this is building sway and its compensation.  Based on the
% design and construction of a building, it will typically have a single,
% dominate mode.

% DVA is useful if you can not get away from the resonance frequency;  can
% not tune a 2 DOF system when the resonance frequency is away from the
% forcing frequency.

% DVA introduces two resonances for the original resonance.  These new
% resonances will produce greater motion at the new resonance frequencies.



%% Environment

close all; clear; clc;
% restoredefaultpath;

addpath( genpath( './00 Support' ), '-begin' );

set( groot, 'DefaultFigurePosition', [ 100  750   750  500 ] ); % x, y, width, height

set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
set( 0, 'DefaultFigureWindowStyle', 'normal' );
set( 0, 'DefaultLineLineWidth', 0.8 );
set( 0, 'DefaultTextInterpreter', 'Latex' );

format ShortG;

pause( 1 );



%% Define Anonymous Functions

h_rpm_to_f = @( rpm )  ( rpm * 0.10471 ) / ( 2 * pi );
h_rpm_to_w = @( rpm )  ( rpm * 0.10471 );

h_f_to_rpm = @( f )  ( f*2*pi ) / 0.10471;

h_w_to_f = @( w )  w/(2*pi);
h_w_to_rpm = @( w )  h_f_to_rpm( w*2*pi );


h_force = @( force_mass, rotation_speed, load_distance )  force_mass .* rotation_speed.^2 .*
    load_distance;



%% Washing Machine Information
```

```
machine.mass = 100;  % kg
machine.rotation_speed = 31.4;  % radians\s or 300 rotations per minute (RPM)
machine.load_mass = 10;  % kg
machine.static_displacement = 1e-2;  % m



%% Load Force Information

dynamic_force.mass = 1;  % kg
dynamic_force.radius = 0.5;  % m

% Maximum force applied to foundation is 100 N.



%% Dynamic Vibration Absorber (DVA)

maximum_allowable_displacement = 1e-2;  % m
    ks = ( machine.load_mass * 9.81) / maximum_allowable_displacement;  % 9,810 N\m


m = 110;  % kg;  washing machine mass and load mass
k = ks;  % 9,810 N\m

wo = sqrt( k / m);  % 9.4 radians\s
    fo = wo/(2*pi);  % 1.5 Hz


m_dva = 15;  % kg
k_dva = 1e5;  % N\m

wo_dva = sqrt( k_dva / m_dva );  % 81.7 radians\s
    fo_dva = wo_dva/(2*pi);  % 13.0 Hz



%% Blocked Frequencies (Equation 9.1 of the Notes)

m1 = m_dva;
k1 = 0;  % Spring above DVA mass;  does not exist;  set to zero.

m2 = m;
k2 = k;

k3 = k_dva;


w1 = sqrt( ( k1 + k3 ) / m1 );  % 81.7 radians\s;  smaller than the system resonant frequency.
    f1 = w1/(2*pi);  % 13 Hz

w2 = sqrt( ( k2 + k3 ) / m2 );  % 31.6 radians\s;  greater than the system resonant frequency.
    f2 = w2/(2*pi);  % 5.0 Hz



%% Coupled Frequencies (Equation 9.15 of the Notes)

mu_4 = ( k3^2 / (m1*m2) );  % 6.1 (kg\s)^4
    mu = ( mu_4 )^0.25;  % 49.1 kg\s

w(1) = 0.5*( ( w1^2 + w2^2 )  +  sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
w(2) = 0.5*( ( w1^2 + w2^2 )  -  sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
    w = sqrt( w );


w_minus = w(2);  % 8.9 radians\s  (lower than 31.6 and 81.7 radians\s)
    f_minus = w(2)/(2*pi);  % 1.4 Hz
%
w_plus = w(1);  % 87.1 radians\s (greater than 31.6 and 81.7 radians\s)
    f_plus = w(1)/(2*pi);  % 13.9 Hz


% The blocked frequencies are always between the lower and higher coupled frequencies.
```

```matlab
%% Mode Shapes (Equations 9.18 and 9.19 of the Notes)

phi_plus = [ ...
     1; ...
     -(1/mu^2)*sqrt(m1/m2)*(w_plus^2 - w1^2), ...
     ];
%
%    1
%    -0.14


phi_minus = [ ...
     1; ...
     (1/mu^2)*sqrt(m1/m2)*(w1^2 - w_minus^2), ...
     ];
%
%    1
%    0.99



%% Admittance Using Equations from DVA Lecture

MAXIMUM_RPM = 3e3;

F0 = 1;

w = 0:0.01:h_rpm_to_w( MAXIMUM_RPM );
     f = w/(2*pi);
          rpm = h_f_to_rpm( f );


m = [ 15 110 ];
k = [  0  k_dva.*(1+0.1*1i)  k2.*(1+0.2*1i)  ];


admittance_modified = F0 ./ ( m(1)*m(2) )*k(2) ./ (( w.^2 - w_plus^2 ) .* ( w.^2 - w_minus^2) );
admittance_original = F0 ./ m(2) * 1 ./ ( w2^2 - w.^2 );


figure( 'Name', 'Admittance - Equations from DVA Lecture' ); ...
     h1 = loglog( rpm, abs( admittance_original ) );  hold on;
     h2 = loglog( rpm, abs( admittance_modified ) );
     h3 = line( [ 300 300 ], [ 1e-6 1e-2 ], 'Color', 'k', 'LineStyle', '--' );
          text( 220, 2e-2, '300 RPM' );
     h4 = line( [ h_f_to_rpm( f_minus )  h_f_to_rpm( f_minus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '-.' );

     h5 = line( [ h_f_to_rpm( f1 )  h_f_to_rpm( f1 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );
     h6 = line( [ h_f_to_rpm( f2 )  h_f_to_rpm( f2 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '--' );

     h7 = line( [ h_f_to_rpm( f_plus )  h_f_to_rpm( f_plus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '--' );    grid on;

     axis( [ 5 3e3  1e-7 1 ] );
          legend( [ h1 h2 h3 h4 h5 h6 h7 ], 'Original System', 'DVA System', 'Operating RPM', ...
               '$w_-$', '$w_1$', ...
               '$w_2$', '$w_+$', ...
               'Location', 'NorthWest', 'Interpreter', 'Latex' );
     xlabel( 'Rotation [RPM]' );  ylabel( 'Admittance [$\frac{m}{N}$]' );



%% Admittance Using nDOF Function

% Damping can be added by using a dampings vector or appending a complex
% value to the respective spring stiffness.  THESE ARE NOT INTERCHANGABLE.


rpm = 0:0.1:MAXIMUM_RPM;
     f = h_rpm_to_f( rpm );
          w = f ./ (2*pi);

FRF = nDOF_direct_solution( m, k, [ 0 0 0 ], f, 'admittance' );  % Symmetric matrix.
%
squeeze( FRF( numel( f ), :, :) );
```

```matlab
%
% (1, 1) - Force on M1 and its associated displacement.
% (2, 2) - Force on M2 and its associated displacement.
%
% (1, 2) - Force on M1 and the displacement of M2.
% (2, 1) - Force on M2 and the displacment on M1.
%
%   These are interchangeable;  the same result.


figure( 'Name', 'Admittance - nDOF Calculated' ); ...
    h1 = loglog( h_f_to_rpm( f ), abs( FRF( :, 1, 1 ) ), 'Color', 'r' );  hold on;
    h2 = line( [ 300 300 ], [ 1e-6 1e-2 ], 'Color', 'k', 'LineStyle', '--' );
        text( 220, 2e-2, '300 RPM' );
    h3 = line( [ h_f_to_rpm( f_minus )  h_f_to_rpm( f_minus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '-.' );

    h4 = line( [ h_f_to_rpm( f1 )  h_f_to_rpm( f1 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );
    h5 = line( [ h_f_to_rpm( f2 )  h_f_to_rpm( f2 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '--' );

    h6 = line( [ h_f_to_rpm( f_plus )  h_f_to_rpm( f_plus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '--' );    grid on;

    axis( [ 2e1 MAXIMUM_RPM  1e-7 1 ] );
        legend( [ h1 h2 h3 h4 h5 h6 ], 'DVA System', 'Operating RPM', ...
            '$w_-$', '$w_1$', ...
            '$w_2$', '$w_+$', ...
            'Location', 'NorthWest', 'Interpreter', 'Latex' );
    xlabel( 'Rotation [RPM]' );  ylabel( 'Admittance [$\frac{m}{N}$]' );


% temp2 = abs( FRF( :, 1, 1 ) );
    % round( temp2( 3001 ), 3, 'significant' )



%% Displacement

dynamic_force.mass = 1;  % kg
dynamic_force.radius = 0.5;  % m


figure( 'Name', 'Displacement' ); ...
    h1 = loglog( rpm, abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius ).', 'Color', 'r' );  hold on;
    h2 = line( [ 300 300 ], [ 1e-7 1e-3 ], 'Color', 'k', 'LineStyle', '--' );  grid on;
        text( 220, 2e-3, '300 RPM' );
    h4 = line( [ h_f_to_rpm( f_minus )  h_f_to_rpm( f_minus ) ], [ 1e-8 1e3 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '-.' );

    h5 = line( [ h_f_to_rpm( f1 )  h_f_to_rpm( f1 ) ], [ 1e-8 1e3 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );
    h6 = line( [ h_f_to_rpm( f2 )  h_f_to_rpm( f2 ) ], [ 1e-8 1e3 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '--' );

    h7 = line( [ h_f_to_rpm( f_plus )  h_f_to_rpm( f_plus ) ], [ 1e-8 1e3 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle', '--' );    grid on;

        legend( [ h1 h2 h3 h4 h5 h6 ], 'DVA System', 'Operating RPM', ...
            '$w_-$', '$w_1$', ...
            '$w_2$', '$w_+$', ...
            'Location', 'NorthWest', 'Interpreter', 'Latex' );

    xlabel( 'Rotation [RPM]' );  ylabel( 'Displacment [m]' );
    axis( [ 2e1 MAXIMUM_RPM  1e-8 1 ] );


% temp2 = abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius ).';
    % round( temp2( 3001 ), 3, 'significant' )



%% Ground Force

dynamic_force.mass = 1;  % kg
dynamic_force.radius = 0.5;  % m
```

```
figure( 'Name', 'Ground Force' ); ...
    h1 = loglog( rpm, abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius
         ).'.*k2, 'Color', 'k' );  hold on;
    h2 = line( [ 300 300 ], [ 1e-4 1e-0 ], 'Color', 'k', 'LineStyle', '--' );  grid on;
        text( 220, 2e-0, '300 RPM' );
    h3 = line( [ 6 MAXIMUM_RPM ], [ 100 100 ], 'Color', 'r' );
        legend( [ h1 h2 h3 ], 'Ground Force', 'Operating RPM', 'Maximum Floor Load', 'Location',
            'SouthEast' );
    xlabel( 'Rotation [RPM]' );  ylabel( ' Force [N]' );
    axis( [ 2e1 MAXIMUM_RPM  1e-5 1e3 ] );


temp2 = abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius ).'*k2;
    % round( temp2( 3001 ), 3, 'significant' )



%% Clean-up

if ( ~isempty( findobj( 'Type', 'figure' ) ) )
    monitors = get( 0, 'MonitorPositions' );
        if ( size( monitors, 1 ) == 1 )
            autoArrangeFigures( 3, 4, 1 );
        elseif ( 1 < size( monitors, 1 ) )
            autoArrangeFigures( 2, 2, 2 );
        end
end

fprintf( 1, '\n\n\n*** Processing Complete ***\n\n\n' );



%% Reference(s)
```

4 Appendix - Matlab Code for Problem 4

```
%% Synopsis

% Problem 4 — Washing Machine Abuse Testing



%% Environment

close all; clear; clc;
% restoredefaultpath;

addpath( genpath( './00 Support' ), '-begin' );

set( groot, 'DefaultFigurePosition', [ 100  750    750  500 ] );  % x, y, width, height

set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
set( 0, 'DefaultFigureWindowStyle', 'normal' );
set( 0, 'DefaultLineLineWidth', 0.8 );
set( 0, 'DefaultTextInterpreter', 'Latex' );

format ShortG;

pause( 1 );



%% Define Anonymous Functions

h_unit_impulse = @( washing_machine_mass, wd, wo, epsilon, t )  ( 1./( washing_machine_mass.*wd) )  .*  exp( -wo.*epsilon.*t ) .* sin( wd.*t );  % mvo = 1 kg\(m s)



%% Impulse Signal

time_step = 1e-4;  % s
net_time = 10;  % s
    time_indices = 0:time_step:( net_time - time_step );

impulse = zeros( size( time_indices ) );
    impulse( 4/time_step ) = 1./time_step;
        brick_impulse = 5 .* impulse;



%% 1 DOF System Impulse Function

ks = 9810;  % N\m
    wo = sqrt( ks / 110 );  % 9.4 radians\s
        fo = wo / (2*pi);  % 1.5 Hz

epsilon = 0.25;
    C = epsilon * 2*sqrt( ks * 110 );  % 519.4 kg\s

wd = wo * sqrt( 1 - epsilon^2 );  % 9.1 radians\s
    fd = wd / (2*pi);  % 1.46 Hz



%% 1 DOF System Impulse Response

impulse_response = h_unit_impulse( 100, wd, wo, epsilon, time_indices );

impulse_response_at_4 = conv( impulse, impulse_response ) * time_step;
    time_indices_at_4 = ( 0:1:( numel( impulse_response_at_4 ) - 1 ) ) .* time_step;
%
% figure( 'Name', '1 DOF System Impulse Response' ); ...
%     yyaxis left; ...
%         plot( time_indices, impulse ); grid on;
%         ylabel( 'Force [N]' );
%         ylim( 'auto' );
%     yyaxis right; ...
```

```
%         plot( time_indices_at_4, impulse_response_at_4 );  hold on;
%         ylabel( 'Admittance [$\frac{m}{N}$]' );
%         ylim( 'auto' );
%     xlabel( 'Time [s]' );
%     xlim( [ 0 10 ] );



%% Combined Response

sinusoid_input = 493.5 * sin( 2 * pi * 5 * time_indices );
    sinusoid_input( 1:1:( 1 / time_step ) ) = 0;
    sinusoid_input( ( 8 / time_step ):1:( 10 / time_step ) ) = 0;


joint_signal = sinusoid_input + brick_impulse;


response_to_combined_forcing = conv( impulse_response, joint_signal ) .* time_step;
    time_indices_sinusoidal_forcing = ( 0:1:( numel( response_to_combined_forcing ) - 1 ) ) .* time_step;
%
figure( 'Name', '1 DOF Combined Forcing Response' ); ...
    h1 = plot( time_indices_sinusoidal_forcing, response_to_combined_forcing );  hold on;
    %
    h2 = line( [ 1 1 ], [ -2e-2 2e-2 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );

    h3 = line( [ 4 4 ], [ -2e-2 2e-2 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '--' );
    h4 = line( [ 6.3 6.3 ], [ -2e-2 2e-2 ], 'Color', 'r', 'LineStyle', '--' );
    %
    h5 = line( [ 8 8 ], [ -2e-2 2e-2 ], 'Color', 'r', 'LineStyle', '-.' );  grid on;

    h6 = line( [4 8 ], [ 5.42e-3 5.42e-3 ], 'LineStyle', ':', 'Color', 'k' );

        legend( [ h1 h2 h3 h4 h5 h6 ], 'Response', 'Start Sinusoidal Forcing', ...
            'Brick Inserted', 'End of Brick Response', ...
            'Stop Sinusoidal Forcing', 'Impulse Response Measure', ...
            'Location', 'NorthEast', 'Interpreter', 'Latex' );

    xlabel( 'Time [s]' );  ylabel( 'Displacement [m]' );
    axis( [ 0 12    -2e-2 2e-2  ] );

return

%% Acceleration

velocity = movingslope( response_to_combined_forcing );
velocity2 = gradient( response_to_combined_forcing );
%
figure( 'Name', 'Velocity' ); ...
    plot( time_indices_sinusoidal_forcing, velocity );  hold on;
    plot( time_indices_sinusoidal_forcing, velocity2, 'LineStyle', '--' );  grid on;
    xlabel( 'Time [s]' );  ylabel( 'Velocity [$\frac{m}{s}$]' );
    % xlim( [ 0 2.5 ] );


acceleration = movingslope( velocity );
acceleration2 = gradient( velocity2 );
%
figure( 'Name', 'Acceleration' ); ...
    plot( time_indices_sinusoidal_forcing, acceleration );  hold on;
    plot( time_indices_sinusoidal_forcing, acceleration2, 'LineStyle', '--' );  grid on;
    xlabel( 'Time [s]' );  ylabel( 'Acceleration [$\frac{m}{s^2}$]' );
    % xlim( [ 0 2.5 ] );



%% Verify

close all;

t = time_indices_sinusoidal_forcing;
    x = sin( 2 * pi * 5 * t );

time_step_verify = t(2) - t(1);

figure( ); ...
    plot( t, x );  hold on;
```

```
    % plot( t(1:1:end-1), diff( x ) / time_step_verify );
    plot( t, gradient( x ) / time_step_verify );
    grid on;
    % axis( [ 0 1  -1.2 1.2 ] );



%% Plot Difference

% figure( 'Name', '1 DOF Combined Forcing Response' ); ...  % Figure 7
%     plot( time_indices_sinusoidal_forcing, response_to_combined_forcing -
    response_to_sinusoidal_forcing );  grid on;
%     xlabel( 'Time [s]' );  ylabel( 'Displacement [m]' );
%     % xlim( [ 0 2.5 ] );



%% Clean-up

if ( ~isempty( findobj( 'Type', 'figure' ) ) )
    monitors = get( 0, 'MonitorPositions' );
        if ( size( monitors, 1 ) == 1 )
            autoArrangeFigures( 3, 4, 1 );
        elseif ( 1 < size( monitors, 1 ) )
            autoArrangeFigures( 2, 2, 2 );
        end
end

fprintf( 1, '\n\n\n*** Processing Complete ***\n\n\n' );



%% Reference(s)
```